Assumptions derive themselves: a member of Gamma is derivable from Gamma

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: verbatim including the !!{formula} token; the bracket title '[Reflexivity]' is the author's own optional argument.

Editorial note. The label prop:reflexivity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Definition of derivation from an assumption set. Earlier in the same chapter: a one-line sequence is a derivation when its line is an axiom or a member of the assumption set. The proof exhibits the one-line derivation and checks the membership clause.

Proposition 0.1 (Reflexivity)**.** *If* $!A \in \Gamma$*, then* $\Gamma \vdash !A$*.*

Proof. The !!formula $!A$ by itself is !!aderivation of $!A$ from Γ. □